Exp-5

Develop a document Question Answering system using Retrieval-Augmented Generation.(RAG)

Aim:

To build a production-grade **Document Question-Answering system** using **RAG** architecture. The system will programmatically ingest a text database, retrieve the top-k most contextually relevant documents using vector search, construct an optimized prompt payload, and leverage an LLM to generate an accurate answer strictly grounded in the retrieved sources.

Algorithm:

1. **Document Chunking & Storage:** Convert your document corpus into dense embeddings using Sentence Transformers and store them in a structural vector space
2. **User Query Input:** Accept an unstructured question from the user.
3. **Semantic Retrieval (The "R" in RAG):** Vectorize the question and search the vector database to pull the top-k closest document text fragments.
4. **Context Augmentation (The "A" in RAG):** Concatenate the pulled document strings into a single text block labeled as Context.
5. **Prompt Engineering:** Inject the Context and the user's Question into a system prompt wrapper that explicitly instructs the LLM not to make things up (hallucinate).
6. **Generation (The "G" in RAG):** Feed the formatted prompt payload to an LLM (e.g., Google Gemini, OpenAI GPT, or a local Hugging Face model) to extract and output a natural language answer.

Code:

```
import numpy as np

import faiss

from sentence_transformers import SentenceTransformer

from google import genai

from google.genai import types

client = genai.Client()

knowledge_base = [

    "Project Apollo was NASA's third human spaceflight program, which succeeded in landing the first humans on the Moon in 1969.",
```

```
    "The Artemis program is a NASA-led human spaceflight initiative intending to re-establish a human presence on the Moon by the mid-2020s.",
    "Mars is the fourth planet from the Sun and is often called the 'Red Planet' due to iron oxide prevalent on its surface.",
    "Voyager 1 is a space probe launched by NASA in 1977 to study the outer Solar System and interstellar space.",
    "The James Webb Space Telescope (JWST) is primarily designed to conduct infrared astronomy, launching successfully in December 2021."
]

print("--- Step 1: Embedding Knowledge Base ---")
embedding_model = SentenceTransformer('all-MiniLM-L6-v2')
dimension = 384
doc_embeddings = embedding_model.encode(knowledge_base)
doc_embeddings = np.array(doc_embeddings).astype('float32')
index = faiss.IndexFlatL2(dimension)
index.add(doc_embeddings)
print(f"Indexed {index.ntotal} source documents successfully.\n")

user_question = "What is NASA's current program to go back to the Moon and when do they plan to do it?"
print(f"User Question: '{user_question}'")

print("\n--- Step 2: Retrieving Relevant Context ---")
query_emb = embedding_model.encode([user_question]).astype('float32')
distances, indices = index.search(query_emb, k=2)
retrieved_chunks = [knowledge_base[idx] for idx in indices[0]]
context_block = "\n".join([f"- {doc}" for doc in retrieved_chunks])
print("Retrieved Context Chunks:")
print(context_block)

print("\n--- Step 3: Generating Grounded Answer (RAG) ---")
rag_prompt = f"""
You are an expert Q&A assistant. Use ONLY the provided Context block to answer the Question.
```

```
If the answer cannot be found in the Context, say "I do not have enough information."

Context:

{context_block}

Question: {user_question}

Answer:

"""

response = client.models.generate_content(

    model='gemini-2.5-flash',

    contents=rag_prompt,

    config=types.GenerateContentConfig(

        temperature=0.0 # Set to 0 to minimize creativity and strictly use context

    )

)

print("\nFinal System Answer:")

print(response.text)
```

Output:

```
--- Step 1: Embedding Knowledge Base ---

Indexed 5 source documents successfully.

User Question: 'What is NASA's current program to go back to the Moon and when do they plan to do it?'

--- Step 2: Retrieving Relevant Context ---

Retrieved Context Chunks:

- The Artemis program is a NASA-led human spaceflight initiative intending to re-establish a human presence on the Moon by the mid-2020s.

- Project Apollo was NASA's third human spaceflight program, which succeeded in landing the first humans on the Moon in 1969.

--- Step 3: Generating Grounded Answer (RAG) ---
```